

Scale-across Architecture and Solutions

SRv6-based Adaptive Load Balancing for AI DCI: A Granularity Progression

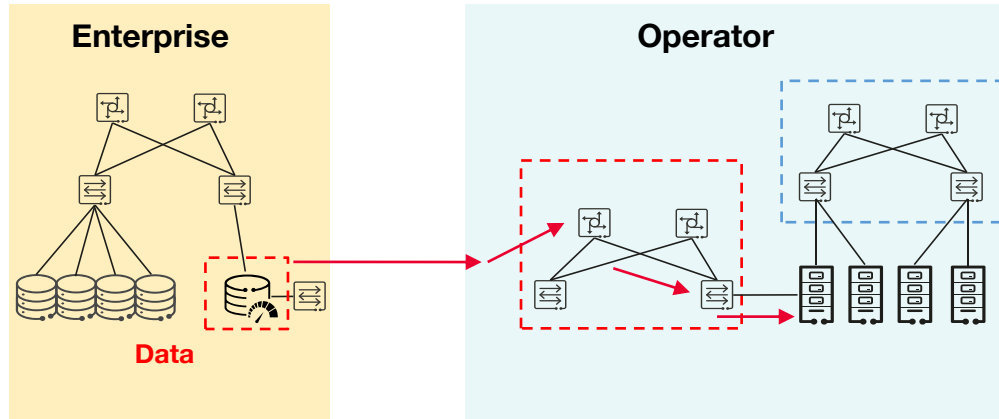
draft-III-srv6ops-dci-srv6-lb-00 | draft-II-idr-flowspec-redirect-sidlist-01
draft-III-idr-flowspec-filter-qp-01 | draft-yang-6man-wide-area-packet-spraying-00

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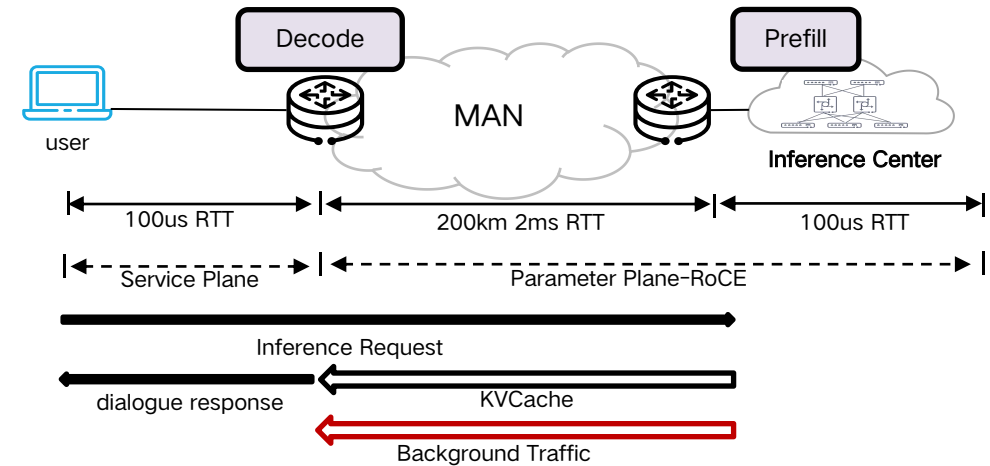
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Problem: AI DCI Elephant Flows over WAN

Why DCI Traffic is Changing



Storage-Compute Separation



PD Separation Deployment

Traffic Characteristics & Challenges

Elephant flows: Gbps+ sustained throughput, hours-to-days duration, bound to stable QP identifiers, strict loss sensitivity

ECMP failure: 5-tuple hash provides insufficient entropy; multiple elephants collide on same path → persistent hotspots

Bandwidth ceiling: Single flow bandwidth can exceed one port's capacity

Solution: Three Levels of Spraying Granularity

Level 1 — Flow-level (5-tuple → SL)



SRv6 Policy with multiple Segment Lists; controller-driven closed loop steers flows between SLs
Scheduling unit = one flow (5-tuple)
Bottleneck: one 5-tuple flow may carry multiple QP elephants; granularity too coarse

Level 2 — QP-level (per-QP sub-flow)



Flowspec Destination-QP matching splits flow into per-QP sub-flows; each steered independently
Scheduling unit = one QP
Bottleneck: a single QP can still be Gbps+;

Level 3 — Packet-group-level (Wide-area Packet Spraying)

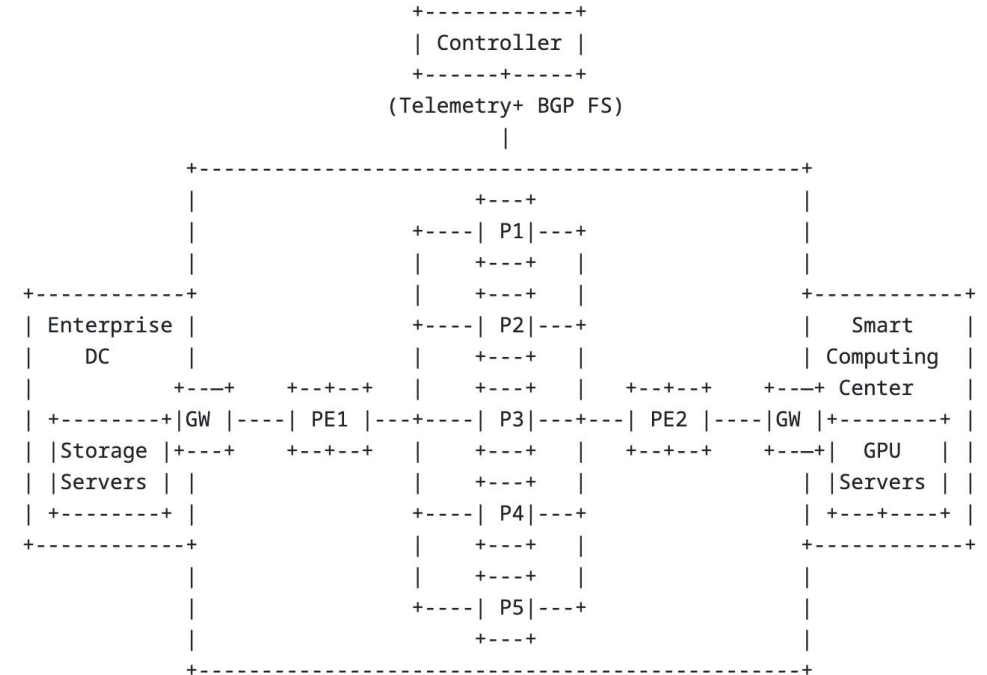


Wide-area Packet Spraying: partition flow into packet groups; spray groups across paths
Scheduling unit = one packet group
Drafts: L1 = dci-srv6-lb + redirect-sidlist | L2 = filter-qp | L3 = wide-area-packet-spraying

Level 1: Flow-level LB — SRv6 Multi-SL Architecture

Architecture

Multiple SRv6 paths between source and destination split high-bandwidth traffic over distinct SLs for parallel transmission (see figure). The controller dynamically distributes traffic according to real-time link load.



Controller-driven Closed Loop

- **Monitoring:** GW identifies elephant flows, reports TOP-N flow info via Telemetry
- **Decision:** Controller subscribes to per-SL link utilization from SRv6 path routers (PE/P); detects SL-level imbalance
- **Enforcement:** Adjust traffic via BGP Flowspec — by 5-tuple (coarse) or by 5-tuple + QP (fine) → redirect to target SL

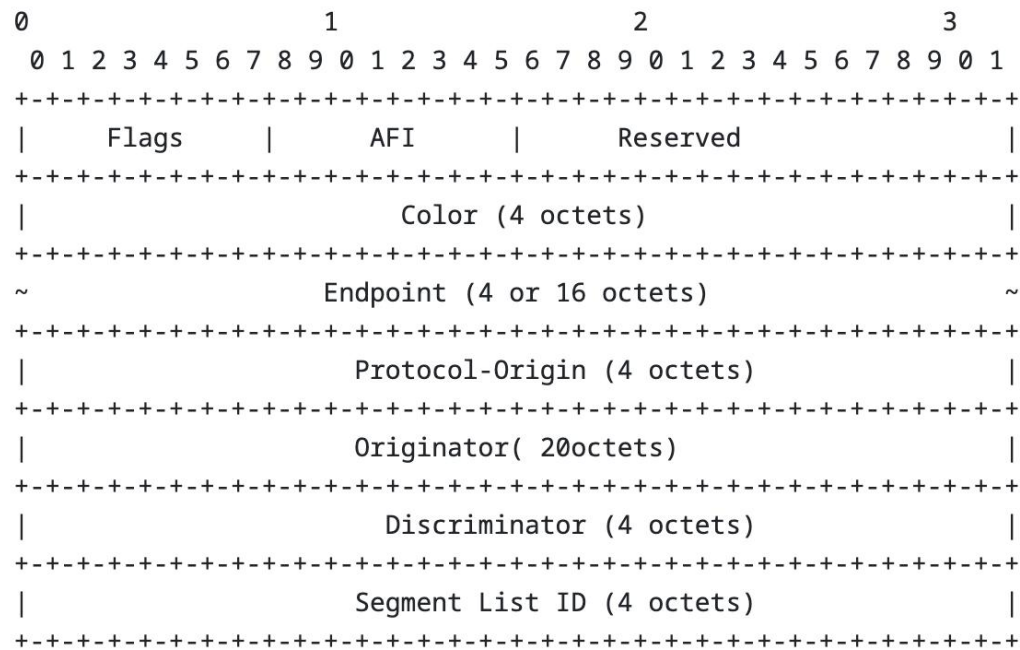
Level 1: Flow-level Enforcement — Flowspec Redirect to SL Gap

Existing Flowspec redirect can only steer into an SR Policy — cannot target a specific Segment List within the Policy

Controller needs SL-level precision for fine-grained load rebalancing across parallel WAN paths

Extension: Redirect to Segment List Action

The Redirect to Segment List action is encoded as a single FSv2 Action SubTLV.



- Procedure:
1. Resolve the SR Policy using <Color, Endpoint>.
 2. Locate the candidate path within that SR Policy that exactly matches <Protocol-Origin, Originator, Discriminator>.
 3. Verify that the candidate path is the active candidate path of the SR Policy (as in [RFC9256] Section 2.9) .
 4. Locate the segment list within that candidate path whose Segment List ID matches the value in the SubTLV.
 5. Verify that the segment list is valid as per [RFC9256] Section 5.

Level 2: QP-level Spraying — Flowspec QP Matching

Why Flow-level is Not Enough

One 5-tuple flow may carry multiple QPs, each being a Gbps+ elephant
Standard 5-tuple Flowspec cannot distinguish QPs sharing the same addresses and ports

Extension: Destination-QP Component

Component:

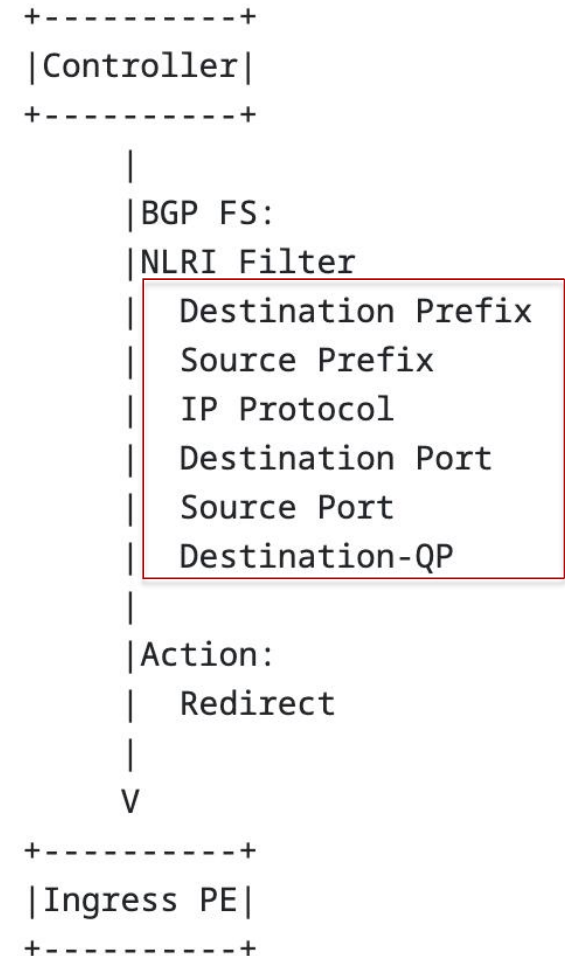
New Flowspec component type: Destination-QP

Matching:

5-tuple + Destination-QP → per-QP sub-flow matching

Combination:

QP matching (who to steer) + SL redirect (where to steer) = complete QP-level scheduling



Level 3: Packet-group Spraying — WPS (1/2)

WPS--Wide-area Packet Spraying

Why QP-level is Not Enough

- Hash collisions and load polarization
- Neither flow-level nor QP-level scheduling can distribute one flow across multiple links
- Need to break the "flow" boundary — spray at finer granularity to aggregate link bandwidth

Overview of Operation

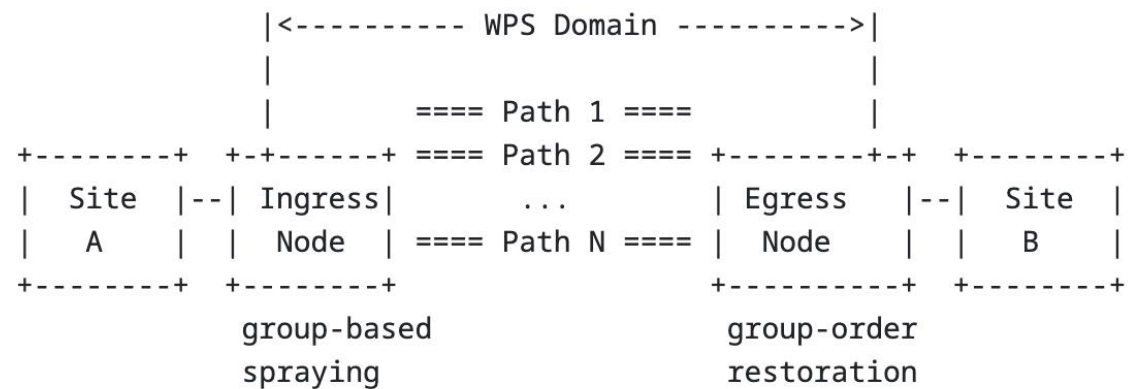
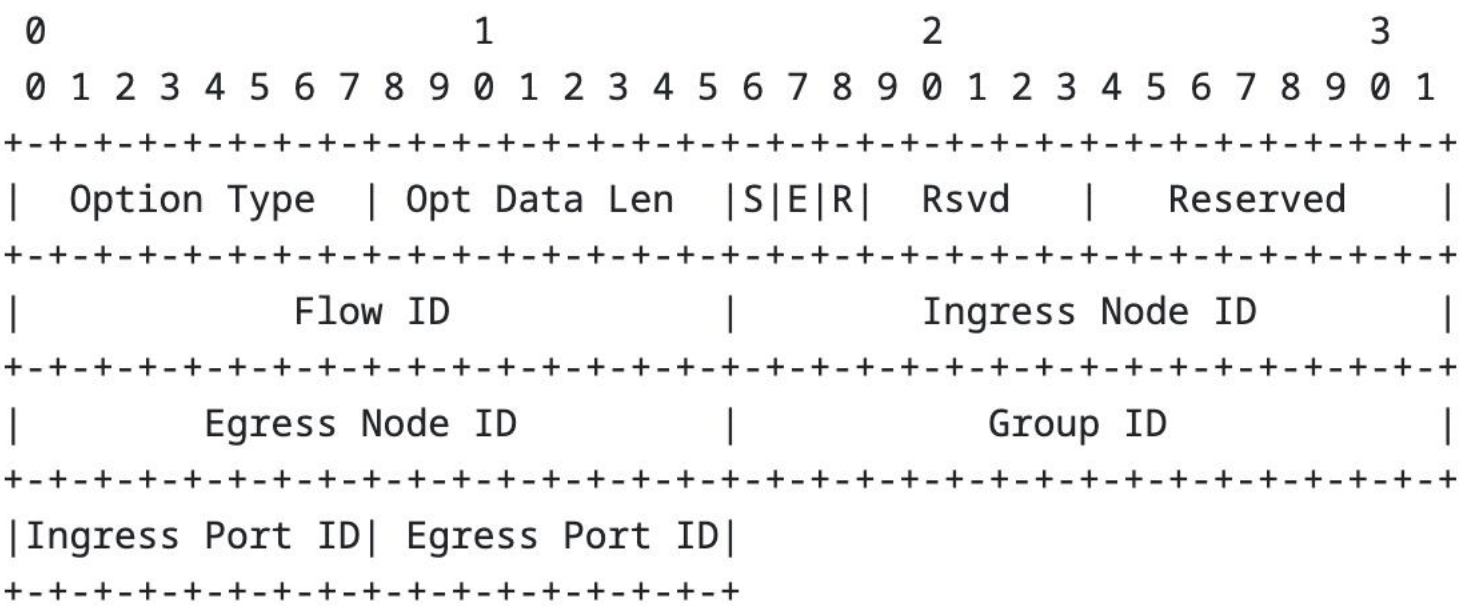


Figure 1: WPS Reference Architecture

- Flow Classification
- Packet Group Partitioning
- Encapsulation & Path Scheduling
- Egress Reassembly Forwarding

Level 3: Packet-group Spraying — WPS (2/2)

Option Format



S bit: marks the start of a packet group

E bit: marks the end of a packet group

R bit: indicates whether the egress node needs to perform reordering

Flow ID
Ingress ID
Egress ID
Group ID

Uniquely identify each flow and its group sequence

Figure 2: WPS Option Format

Destination Options Header-----RECOMMENDED

Hop-by-Hop Options Header

Summary & Next Steps

Granularity Progression — Complementary, Not Replacement

Level	Unit	Mechanism	Draft(s)
Flow	5-tuple flow	SRv6 multi-SL + Flowspec redirect to SL	draft-III-srv6ops-dci-srv6-lb-00 draft-II-idr-flowspec-redirect-sidlist-01
QP	per-QP sub-flow	Flowspec Destination-QP matching	draft-III-idr-flowspec-filter-qp-01
Packet Group	packet group	WPS: group-based spraying + egress reorder	draft-yang-6man-wide-area-packet-spraying-00

Three levels solve different problems

- L1: Flow-level: balance across flows (controller closed loop)
- L2: QP-level: balance sub-flows within one 5-tuple (per-QP steering)
- L3: Packet-group: aggregate bandwidth for a single flow (break port capacity ceiling)

Next Steps

Feedback on architecture and granularity progression approach
Details: SRV6OPS(multi SL) | IDR (QP component + SL redirect) | 6MAN(WPS IPv6 option)